Plant Intelligence Lab

Genomic Prediction, Phenotype Forecasting
& AI-Assisted Biological Decision Systems

Technical Architecture and Public Research Document

Can we predict a plant phenotype from genomic information, environmental variables and early biological observations, while quantifying uncertainty?

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Public technical document • Computational plant biotechnology • 2026

Plant Intelligence Lab

A technical narrative for predictive plant biotechnology

Purpose. This document describes the public scientific architecture of Plant Intelligence Lab: a reproducible computational framework for genomic prediction, phenotype forecasting, uncertainty-aware machine learning, and AI-assisted biological decision support. It is written for researchers, quantitative scientists, biotechnology teams, technical founders, and practitioners interested in translating biological data into better predictions and better experiments.

The document is intentionally public-facing. It describes the scientific questions, model classes, validation logic, and industrial relevance of the project without exposing private business strategy or proprietary implementation details.

Public repository: `https://github.com/rolffcoelho-bravo/plant-intelligence-lab`

This project uses public biological data for reproducible research and benchmarking. Results obtained on public datasets are not presented as validated performance on proprietary industrial processes. Prediction is not biological causation, and uncertainty is treated as part of the output rather than an afterthought.

Contents

The Question at the Center

Biological systems are rich in information, but that information arrives in fragments. A genome describes inherited possibility. An environment changes the expression of that possibility. A treatment perturbs the system. Early observations reveal a trajectory that has not yet reached its outcome. The challenge is to combine these pieces before time has already answered the question for us.

Plant Intelligence Lab begins with one deceptively simple problem:

Can we predict a plant phenotype from genomic information, environmental variables and early biological observations, while quantifying uncertainty?

The general forecasting target is

$$\widehat{Y}_{t+h} = f(G, P, E, X_t)\,, \tag{1}$$

where

- $\widehat{Y}_{t+h}$ is a predicted biological outcome at a future horizon $t + h$;
- G represents genomic information;
- P represents protocol or treatment information;
- E represents environmental information;
- X_t contains biological observations available up to time t.

Equation (1) is not a claim that every dataset contains every component. It is the architecture toward which the project develops. Some experiments may begin with genotype and phenotype only. Others can add treatment, environment, imaging, longitudinal observations, or operational variables as the evidence base grows.

The key shift is conceptual: the repository is not designed around a single algorithm. It is designed around a *decision problem*. Models are useful only insofar as they make future biological outcomes more predictable, more interpretable, or more actionable.

Public Data Foundation

The first empirical layer is built from public plant-genomics resources so that the work can be inspected, reproduced, challenged, and improved.

The 1001 Genomes Project reports a major 2016 phase containing detailed analysis of 1,135 *Arabidopsis thaliana* genomes. AraPheno provides public phenotype data associated with Arabidopsis accessions. A particularly relevant AraPheno study subjects 170 natural accessions to two variants of a shoot-regeneration protocol and records regenerated shoot number and related *in vitro* traits. These resources create a rare public setting where genetic variation, experimental treatment, and regeneration phenotypes can be studied together.

Why this matters. A generic crop-yield benchmark can demonstrate machine learning. A regeneration dataset can demonstrate something closer to plant biotechnology itself: genotype-dependent response under controlled experimental intervention.

The public data strategy is deliberately conservative. Raw data provenance, accession matching, phenotype definitions, missingness, and licensing are treated as scientific inputs, not clerical details. The predictive system is only as credible as the biological meaning of the data supplied to it.

Step 1 - Establish the Quantitative-Genetics Baseline

Before asking whether a nonlinear machine-learning model can improve prediction, the repository establishes a classical reference point. In genomic prediction, a natural benchmark is a linear mixed model:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\mathbf{u} + \boldsymbol{\varepsilon}, \tag{2}$$

with

$$\mathbf{u} \sim \mathcal{N}\left(\mathbf{0}, \mathbf{K}\sigma_g^2\right), \qquad \boldsymbol{\varepsilon} \sim \mathcal{N}\left(\mathbf{0}, \mathbf{I}\sigma_e^2\right). \tag{3}$$

Here, $\mathbf{K}$ is a genomic relationship matrix, σ_g^2 captures genomic variance, and σ_e^2 captures residual variance. In a GBLUP-style formulation, genomic similarity becomes part of the covariance structure through which breeding values or genomic effects are estimated.

This baseline serves two purposes. Scientifically, it anchors the analysis in quantitative genetics. Computationally, it prevents the project from confusing model complexity with model quality. If a sophisticated learner cannot outperform or complement a well-specified baseline under honest validation, that result is informative.

What is evaluated

The baseline should answer:

- How much predictive signal is available from genomic relatedness alone?
- How stable are predictions across held-out accessions?
- How much of the variation is captured by genotype relative to treatment and residual variation?
- Does the genomic relationship structure reveal clusters that must be respected during validation?

The `sommer` ecosystem is relevant here because it provides flexible mixed-model machinery for genomic prediction and variance structures, including additive, dominance, and epistatic relationship components.

Step 2 - Enter the High-Dimensional Regime

Genomics creates an inversion of the ordinary data problem. There may be thousands, hundreds of thousands, or millions of molecular markers, while the number of observed plants remains comparatively small. The statistical regime is therefore

$$p \gg n, \tag{4}$$

where p denotes genomic features and n denotes observed accessions or plants.

When $p \gg n$, an algorithm can memorize structure that will not survive contact with unseen biology. The task is therefore not to obtain an impressive in-sample score. The task is to identify signal that survives separation from the data on which the model was trained.

Model family

The public benchmark can compare a disciplined set of models:

- Elastic Net and other regularized linear models;
- Random Forest;
- gradient boosting, including XGBoost or LightGBM;
- kernel methods;
- carefully regularized neural networks where sample size supports their use.

The comparison is not a leaderboard for the prettiest R^2. It is a test of generalization, calibration, biological plausibility, and computational stability.

Performance metrics

For continuous phenotypes, core error measures include

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \widehat{y}_i)^2}, \tag{5}$$

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}\left|y_i - \widehat{y}_i\right|, \tag{6}$$

and predictive correlation

$$\rho(y, \widehat{y}) . \tag{7}$$

R^2 can be reported as a familiar summary, but it is not allowed to become the sole definition of success. In biological forecasting, the quality of the split, the novelty of the held-out samples, and the calibration of uncertainty can matter as much as the headline score.

Step 3 - Validate on Unseen Genotypes

A random train-test split can look rigorous while quietly leaking biological similarity across the boundary. If closely related accessions appear in both training and test sets, the model may be evaluated on a softer problem than the one an industrial or research deployment would actually face.

Plant Intelligence Lab therefore treats genotype-aware validation as a first-class design requirement.

Validation principle. The test set should represent the biological generalization claim. If the claim is prediction for unseen genotypes, validation must hold out genotypes in a way that limits information leakage through genetic relatedness.

Possible strategies include grouped cross-validation by accession or genetic cluster, relationship-aware folds, and deliberately distant holdouts. The exact protocol depends on the available genomic structure, but the principle remains fixed: the evaluation must be harder than memorizing kinship.

This distinction is central to the repo's credibility. A lower but honest estimate of out-of-sample performance is more valuable than an inflated result that disappears when the model meets truly new material.

Step 4 - Model Genotype-by-Environment Response

A genotype does not perform in a vacuum. Biological performance can change with temperature, water, soil, light, treatment conditions, and other environmental factors. A standard decomposition is

$$Y_{ij} = \mu + G_i + E_j + (G \times E)_{ij} + \varepsilon_{ij}, \tag{8}$$

where G_i is a genotype effect, E_j is an environment effect, and $(G \times E)_{ij}$ captures their interaction.

A flexible predictive extension is

$$\widehat{Y} = f(G, E, G \times E)\,. \tag{9}$$

The practical question is not merely which genotype performs best. It is which genotype performs best *under which conditions*, with what expected variation, and with what confidence.

Industrial interpretation. A biological system that predicts only average performance may be too blunt for operational use. A system that estimates conditional performance across environments begins to support selection, allocation, and risk-aware planning.

The same logic applies to genotype-by-treatment effects. Where protocol or treatment variables are available, the relevant model can expand to $f(G, P, G \times P)$ and eventually to richer interactions among genotype, environment, and protocol.

Step 5 - Forecast the Outcome Before the Experiment Is Finished

Time is an expensive biological variable. A result that becomes obvious only at the end of a long experiment cannot recover the resources already consumed. Early biological forecasting asks whether partial trajectories contain enough information to anticipate the future.

A general formulation is

$$P(Y_T \mid G, P, E, X_{0:t}) \,, \tag{10}$$

where $X_{0:t}$ contains information observed from the beginning of an experiment through time t, and Y_T is a later outcome.

For the regeneration setting, one concrete comparison is

$$G + P + X_{15d} \;\longrightarrow\; \widehat{Y}_{21d} \tag{11}$$

versus

$$G + P \;\longrightarrow\; \widehat{Y}_{21d}. \tag{12}$$

The difference between these models quantifies the incremental forecasting value of early biological information.

If early measurements materially improve prediction, the system becomes more than a static genomic predictor. It becomes a dynamic decision instrument: as observations arrive, forecasts can be updated.

A possible decision interface

A later application layer could summarize a batch or accession with outputs such as:

- predicted probability of successful development;
- prediction interval;
- primary predictive drivers;
- evidence of distribution shift;
- confidence state.

The numbers themselves are not the innovation. The architecture is: a biological process that was previously observed retrospectively becomes monitored prospectively.

Step 6 - Treat Uncertainty as Part of the Prediction

A model that reports only $\widehat{Y}$ compresses a scientific problem into a false point of certainty. Plant Intelligence Lab instead treats uncertainty as a first-class output.

A predictive interval can be represented as

$$P(L_{1-\alpha}(x) \leq Y_{\text{new}} \leq U_{1-\alpha}(x)) \approx 1 - \alpha, \tag{13}$$

where $L_{1-\alpha}(x)$ and $U_{1-\alpha}(x)$ are lower and upper predictive bounds for a new observation.

Candidate methods include bootstrap intervals, conformal prediction, Bayesian predictive distributions, and other calibrated uncertainty procedures appropriate to the chosen estimator and data structure.

Abstention

A mature predictive system should know when it is outside its evidence base. If a genotype, environment, or combination is sufficiently different from training data, the model should be allowed to refuse false precision.

LOW CONFIDENCE - insufficient evidence for reliable prediction

This is not failure. It is scientific discipline encoded into software.

A useful abstention rule can be framed as

$$\text{predict only if } C(x) \geq \tau, \tag{14}$$

where $C(x)$ is a calibrated confidence or support score and τ is a decision threshold. The precise definition of $C(x)$ is model-dependent and must be validated rather than chosen decoratively.

Step 7 - Move from Prediction to Experiment Selection

Prediction asks what is likely to happen. Experimental intelligence asks what should be tested next.

Suppose the experimental budget permits only a small subset of possible genotype-treatment-environment combinations. Rather than choose them randomly, the system can rank candidates by expected improvement, uncertainty, information gain, or a controlled combination of those objectives.

A generic acquisition rule is

$$x^{\star} = \arg\max_{x} \left[\mathbb{E}(\text{Improvement} \mid x) + \lambda\, U(x) \right], \tag{15}$$

where $U(x)$ represents uncertainty or information value and λ controls the exploration-exploitation trade-off.

The public implementation can begin as a transparent active-learning or Bayesian-optimization simulator. The purpose is not to claim autonomous biology. The purpose is to show how a validated predictive model can allocate scarce experimental capacity more intelligently.

> **Decision value.** In research environments where experiments are slow, expensive, or capacity-constrained, the question "what should we test next?" can be economically as important as "what will happen?"

A future recommendation could include the candidate combination, predicted value, uncertainty, and a short explanation of why the experiment was prioritized.

Step 8 - Add GenAI Only After the Quantitative Engine Works

Generative AI is useful here, but not as a substitute for modelling. The architecture places the language model at the interface, not at the center of scientific truth.

$$\boxed{\text{Scientist}} \longleftrightarrow \boxed{\text{GenAI}} \longleftrightarrow \boxed{\text{Models + Database}}$$

The language interface can translate a researcher's question into a structured query against validated data and model outputs. It can summarize evidence, explain predictions, retrieve experimental history, and expose uncertainty in natural language.

What it should not do is invent biological results.

For example, a user might ask which accessions exhibit unusually high predicted regeneration under a particular treatment. The system should answer from the underlying database and model layer, with traceable evidence and uncertainty. The LLM becomes a scientific interface to computation rather than a generator of unsupported conclusions.

Case Study A - In-Vitro Regeneration Intelligence

The first case study is designed around public *Arabidopsis thaliana* resources linking genomic variation to regeneration phenotypes and treatment conditions.

The core empirical structure is:

$$\boxed{\text{Genetic variation}} + \boxed{\text{In-vitro treatment}} \longrightarrow \boxed{\text{Regeneration phenotype}}. \tag{16}$$

The study should answer a sequence of increasingly difficult questions:

1. Can regeneration-related phenotypes be predicted for unseen accessions?
2. How much predictive information is contained in genomic structure?
3. Does treatment information materially improve prediction?
4. Are genotype-by-treatment interactions informative?
5. Do early biological observations add forecast value?
6. How well calibrated is predictive uncertainty?

7. Under what conditions should the model abstain?

This case study is intentionally close to a real biotechnology decision context. It is not merely an exercise in fitting a classifier or regressor to a tabular benchmark.

Case Study B - Genotype-by-Environment Forecasting

The second case study broadens the problem from controlled regeneration to multi-environment prediction. Public wheat data available through the `sommer` ecosystem provide a suitable environment for demonstrating genomic prediction across multiple agroclimatic settings.

The objective is not simply to predict yield. The objective is to test whether the modelling architecture can generalize when environmental context changes and whether uncertainty expands appropriately under harder transfer conditions.

This case study is where the project can demonstrate:

- genomic prediction under environmental heterogeneity;
- genotype-by-environment interaction;
- environment-aware validation;
- distribution-shift diagnostics;
- probabilistic forecasting under changing conditions.

Case Study C - AI-Guided Experiment Selection

The third case study turns validated prediction into experimental prioritization. It constrains the number of experiments that may be run, scores candidate combinations, and compares different selection policies.

The progression is

$$\text{Prediction} \longrightarrow \text{Uncertainty} \longrightarrow \text{Information Value} \longrightarrow \text{Experiment Selection.} \tag{17}$$

The benchmark can compare random selection, exploitation-heavy selection, uncertainty sampling, and a balanced acquisition rule. The result is not a claim that algorithms should replace experimental judgment. It is a test of whether mathematical prioritization can improve the use of scarce experimental capacity.

Repository Architecture

The codebase is organized so that exploratory work, reusable implementation, experiments, results, and public documentation remain distinct.

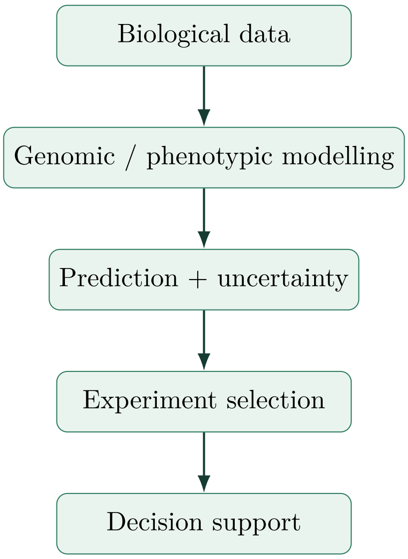
```
plant-intelligence-lab/
|
|-- README.md
|-- LICENSE
|-- CITATION.cff
|-- pyproject.toml
|
|-- data/
|   |-- README.md
|   |-- raw/
|   |-- interim/
|   `-- processed/
|
|-- notebooks/
|   |-- 01_data_discovery.ipynb
|   |-- 02_genomic_structure.ipynb
|   |-- 03_genomic_prediction.ipynb
|   |-- 04_gxe_forecasting.ipynb
|   |-- 05_uncertainty.ipynb
|   `-- 06_active_learning.ipynb
|
|-- src/
|   `-- plant_intelligence/
|       |-- data/
|       |-- genetics/
|       |-- models/
|       |-- forecasting/
|       |-- uncertainty/
|       |-- optimization/
|       `-- explainability/
|
|-- experiments/
|   |-- baselines/
|   |-- ml/
|   `-- genomic/
|
|-- reports/
|   |-- figures/
|   |-- model_cards/
|   `-- results/
|
|-- app/
|   `-- dashboard/
|
|-- tests/
|
`-- docs/
    |-- methodology.md
    |-- biological_context.md
    |-- limitations.md
    `-- transferability.md
```

This structure is deliberately conventional enough to be readable and disciplined enough to scale. Notebooks expose reasoning and exploration. The `src/` package contains reusable code. Experiments preserve model comparisons. Reports hold outputs rather than strategy. Documentation explains the scientific context, methodology, limitations, and transferability.

Industrial Transferability

Public datasets can establish technical competence and empirical behavior. They cannot establish performance on a proprietary biological process that has never been observed. Plant Intelligence Lab therefore treats industrial transfer as a conditional translation problem.

The transferable architecture is:



Potential application domains include genomic selection, plant propagation, regeneration analysis, treatment-response modelling, environment-aware prediction, early outcome forecasting, experimental prioritization, quality control, and scientific data interfaces.

The appropriate industrial application begins by identifying the biological decision, the data already captured, the cost of error, and the value of earlier information. Only then should a modelling architecture be selected.

What Makes the Project Technically Serious

The value of Plant Intelligence Lab does not come from listing many algorithms. It comes from the order in which the problem is attacked.

1. **Biological meaning before modelling.** Phenotypes, accessions, treatments, and environments are defined before code is optimized.
2. **Classical baseline before complexity.** Quantitative genetics establishes the reference point.
3. **Generalization before headline scores.** Validation is designed around unseen biology.
4. **Uncertainty before automation.** A forecast without uncertainty is incomplete.
5. **Abstention before false confidence.** The system is allowed to say that evidence is insufficient.
6. **Prediction before optimization.** Experimental selection is built only after the predictive layer has earned trust.

7. **Quantitative engine before GenAI.** Language models interface with scientific outputs; they do not replace them.

That sequence is the difference between an AI demo and a research system.

Public Development Path

The repository can be developed in a visible technical sequence while keeping public communication focused on evidence rather than internal strategy.

1. Data discovery and accession matching

Verify phenotype definitions, genomic availability, sample intersections, missingness, population structure, and reproducible acquisition.

2. Genomic structure

Characterize relatedness, dimensionality, population structure, and the implications for cross-validation.

3. Genomic prediction

Fit the quantitative-genetics baseline and selected high-dimensional models under genotype-aware validation.

4. Genotype-by-environment forecasting

Add environment and interaction structure where suitable public data permit a real transfer problem.

5. Uncertainty

Calibrate predictive intervals, detect unsupported inputs, and implement abstention criteria.

6. Active learning and experiment selection

Use validated predictions and uncertainty to rank candidate experiments under constrained capacity.

7. Scientific interface

Expose verified data and model outputs through an interpretable, traceable GenAI layer.

A Compact Mathematical View

The full system can be summarized as a sequence of maps.

Observation to forecast

$$(G, P, E, X_t) \mapsto \widehat{Y}_{t+h}. \tag{18}$$

Forecast to calibrated uncertainty

$$\widehat{Y}_{t+h} \mapsto \left(\widehat{Y}_{t+h},\ \mathcal{I}_{1-\alpha}(x),\ C(x)\right), \tag{19}$$

where $\mathcal{I}_{1-\alpha}(x)$ is a predictive interval and $C(x)$ is a support or confidence measure.

Prediction to experiment choice

$$\left(\widehat{Y}(x), U(x), \mathrm{cost}(x)\right) \mapsto x^{\star}. \tag{20}$$

Scientific system to human interface

$$\text{Question} \longrightarrow \text{verified query} \longrightarrow \text{model/data evidence} \longrightarrow \text{explanation.} \tag{21}$$

This is the project's central ambition: not simply to predict a phenotype, but to connect biological evidence to forecast, uncertainty, experimental choice, and interpretable decision support.

Limitations and Scientific Boundaries

A serious public repository must be explicit about what its results do not mean.

- Predictive association does not establish biological causation.
- Small-sample genomic settings can produce unstable estimates and optimistic validation if relatedness is mishandled.
- Public *Arabidopsis* results do not automatically transfer to another species, production system, laboratory, or commercial setting.
- Environment and treatment variables may be incompletely measured.
- Uncertainty intervals depend on assumptions or coverage definitions that must be stated clearly.
- AI-assisted experiment selection is decision support, not autonomous experimental authority.
- GenAI explanations are only as trustworthy as the underlying data, models, retrieval logic, and evidence traceability.

These limits do not weaken the project. They define the conditions under which its results can be believed.

Conclusion - From Biological Data to Better Decisions

The promise of computational plant biotechnology is not that every biological process can be reduced to an algorithm. The promise is narrower, more useful, and more demanding: that some uncertainty can be quantified before time resolves it; that some experiments can be prioritized before capacity is spent; that some genotype-environment relationships can be modelled before selection becomes guesswork; and that scientific evidence can be exposed through systems that remain transparent about what they know and what they do not.

Plant Intelligence Lab is built around that proposition.

Biological Data → Quantitative Genetics → Machine Learning → Forecasting → Uncertainty → Experiment Optimization → Decision Intelligence

The repository is therefore not a collection of fashionable models. It is an attempt to build a coherent predictive architecture around a real biological problem - one in which high-dimensional genomics, statistics, machine learning, uncertainty quantification, optimization, explainability, and GenAI each have a defined role.

The standard is simple to state and difficult to satisfy: predictions should generalize, uncertainty should be visible, experiments should become more informative, and every result should remain reproducible enough to be challenged.

References and Public Data Sources

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3. Lardon, R., Wijnker, E., Keurentjes, J. J. B., and Geelen, D. (2020). *The genetic framework of shoot regeneration in Arabidopsis comprises master regulators and conditional fine-tuning factors.* Communications Biology 3. DOI: 10.1038/s42003-020-01274-9.

4. Covarrubias-Pazaran, G. (2016). *Genome-Assisted Prediction of Quantitative Traits Using the R Package sommer.* PLOS ONE 11(6): e0156744. DOI: 10.1371/journal.pone.0156744.